

Multi-source DINOv2 fMRI foundation model — v1 results

2026-06-29

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This document reports the first complete, leakage-free results of the multi-source DINOv2 fMRI foundation model. It is organised in four parts:

1. **What the 2026-06-14 meeting asked, and how each point was implemented** (including where the implementation deviated from the written plan, and why).
2. The **two ablations** that were run (base vs Fourier).
3. The **evaluation protocol** (leakage-free holdout, 70:30 split, cross-validation).
4. The **per-task results next to every SOTA that reports the same benchmark**.

Executive summary (two messages)

- **GOOD: Demographic / global structure is strong.** On **ABIDE Age** we reach **0.80 accuracy**, above the volumetric peers (SLIM-Brain 0.64, SwiFT 0.62); on **HCP Sex** we reach **0.81 acc / 0.79 F1**, above **LCM** (0.73 F1).
- **WEAK: Fine clinical signal is near chance.** Autism (ABIDE) and early Alzheimer (ADNI NC/MCI, AD/HC) stay around 0.5–0.64, **below** the graph/connectome SOTA.
- The **Fourier** positional encoding **does not help** (base $\geq$ Fourier on every axis that carries signal) -> the retained model is **base**.

1. From the meeting plan to the implementation

The 2026-06-14 meeting (MEETING_SUMMARY_2026-06-14) defined four work items. Below is what was asked and what was actually built — two items were implemented **differently** from the written plan, for concrete technical reasons.

1.1 Multi-source pretraining corpus

Asked: extend HCP-YA-only pretraining with aging cohorts (HCP-Aging, OASIS-3, AOMIC), ~3.4k sessions, to close the age gap with the ADNI downstream target.

Built: a **5-source corpus of 4 627 scans**, all resampled to (T, 45, 54, 45):

Source	Scans	Role
HCP-YA	1 084	young healthy baseline
ABIDE I	1 102	autism + demographic benchmark
OASIS-3	1 197	aging + AD
AOMIC PIOP1/2	434	scanner diversity
ADNI	812	Alzheimer downstream target

Deviation: HCP-Aging access (NDA-controlled, 1–3 months) was not ready, so **ABIDE I and ADNI** were used instead — they also give us the canonical fMRI benchmarks (ABIDE autism, ADNI NC/MCI). Scale (~4.6k) matches the SLIM-Brain target.

1.2 Mixing scans of different length / different TR

Asked (written plan): pad every scan along time to the longest $T = 1200$ (HCP-YA), zero-padding the end; respect a padding mask in the loss.

Built — a different solution (Ariel’s later decision): padding to 1200 would have meant **60–85% zeros** for most datasets, which the masking objective cannot learn from. Instead we do **online temporal harmonisation**:

- every scan is resampled from its native TR to a common **TR = 0.72 s** using `scipy.signal.resample_poly` (we crop the native window **first**, then resample — this keeps the polyphase filter cheap);
- then a fixed window of $T_{\text{fixed}} = 270$ **frames** (~ 194 s) is taken.

This resolves the meeting’s **open question 2 (TR heterogeneity)** by option 1 (resample to a common TR): one token now spans the **same physical duration** across all datasets. Native TRs: HCP 0.72, ABIDE per-site, OASIS 2.2, AOMIC 0.75/2.0, ADNI 3.0.

Patchify (unchanged PatchEmbed3DPlus1D): hierarchical 3D-spatial + 1D-temporal, `temporal_kernel = 10` => $T_{\text{eff}} = 27$, token grid $27 \times 150 = 4050$.

1.3 Augmentation: masking-only

Asked: replace DINOv2 multi-crop (spatial sub-volumes + BeiT block masking) by **1 global + 3 local** full-image crops with **per-token Gaussian masking**, because (a) cropping a sub-volume of a fixed brain is not semantically meaningful, and (b) BeiT block masking is spatially incoherent on the flattened token grid.

Built — the masking part, pragmatically: all crops are the **full volume** (no spatial zoom) with **per-token random masking** (`RandomTokenMaskingGenerator`, ratio sampled in $[0.10, 0.50]$), which replaces BeiT block masking as asked. We **kept DINOv2’s proven 2-global + 3-local self-distillation loss intact** rather than rewriting it to the literal “1 global + 3 local” — the literal form would require rewriting `forward_backward` and was judged too risky for a first iteration. So: **full-image crops + MAE-style per-token masking, original loss**.

1.4 Fourier features for spatial position (the §3 ablation)

What the model needs. Each of the 150 spatial patches must be told *where it sits* in the brain so the transformer can use spatial relationships. By default this is a **learned lookup table** (`pos_spatial`, 150x384): 150 vectors learned from scratch, with **no built-in geometry** — nothing tells the model that patch (0,0,0) is a neighbour of (0,0,1).

What Fourier features do (Tancik et al., NeurIPS 2020). A plain network fed raw coordinates suffers *spectral bias*: it cannot tell apart inputs that are **close together** — it blurs nearby positions into the same output. Fourier features fix this by mapping a coordinate through **cos/sin at many frequencies**: two nearby positions then produce **clearly different** signatures (at high frequency) while staying correlated (at low frequency). In short, they let the network **resolve fine spatial distances** that raw coordinates cannot express.

Built — exactly as the meeting’s Option B, in `dinov2/layers/patch_embed_3d_plus_1d.py`. The encoding is computed in **three steps** that replace the learned table:

1. **Coordinate** — give each of the 150 patches its 3D position in $[-1, 1]$ (`_make_grid_coords`, in the same order the patchify flattens the tokens).
2. **Fourier map** — $\text{gamma}(v) = [\cos(2\pi \cdot Bv), \sin(2\pi \cdot Bv)]$: turn the 3 numbers into 64 (32 cosines + 32 sines), where B is a fixed random frequency matrix.
3. **MLP** — a 2-layer MLP maps the 64 features to `embed_dim = 384`, the per-token size, and this is added to each token as its spatial position.

```
class FourierFeatures3D(nn.Module):          # step 2
    def __init__(self, num_freqs=32, sigma=10.0):
        self.register_buffer('B', torch.randn(num_freqs, 3) * sigma) # FIXED freqs
    def forward(self, pos):                  # (150,3) -> (150,64)
        proj = 2*math.pi * pos @ self.B.t()
        return torch.cat([torch.cos(proj), torch.sin(proj)], dim=-1)
```

A single switch (`get_spatial()`) returns either the learned table (**base**) or this Fourier chain (**fourier**) — that switch is the only difference between the two runs. Hyperparameters: `num_freqs = 32`, `sigma = 10`, B fixed (the meeting’s “fixed vs learnable B ” -> fixed for $v1$). Only the **spatial** position is replaced; temporal and CLS positions stay learned. Toggled by `student.fmri_fourier_pos` (`FOURIER=1`).

1.5 Transfer learning + freeze policy

The transformer is **initialised from the ImageNet DINOv2 checkpoint** (ViT-S/14, 4 register tokens); only the fMRI-specific 3D patchify + positional encoding are random-init. We use freeze policy **B** (`fmri_plus_last_3`): transformer blocks 0–8 are frozen, and the patchify + blocks 9–11 + final norm + heads are trained (~23.3 M trainable params). Both ablations use policy B.

1.6 Leakage (meeting open question 1) — resolved

The meeting left “can we pretrain on the full data without leakage” open. **Resolved (2026-06-28, confirmed by Yoni)**: exclude the **downstream test subjects from SSL pretraining**. `HOLDOUT_DATASETS = (ADNI, ABIDE, OASIS, HCP)` — the held-out 30% of each is removed from pretraining, so the encoder never sees an evaluation subject. This matches the LCM / Brain-JEPA practice. (Full SOTA-exact alignment would pretrain on a disjoint UK Biobank — planned for $v2$ once access is granted.)

2. The two ablations

Both runs share everything in Section 1 and differ by **one factor only** (one-factor-at-a-time design): the spatial positional encoding.

Run	Spatial position	Freeze	Init	Epochs
base	learned table	B (fmri_plus_last_3)	DINOv2 ImageNet	10
fourier	Fourier features of 3D coords	B	DINOv2 ImageNet	10

Each run: ~23 000 iterations, effective batch 16 (micro-batch 2 x grad-accum 8), proportional sampling across the 5 sources, single GPU (~3.5 h).

3. Evaluation protocol

Embedding. For each scan: one embedding = mean CLS token over sliding T_{fixed} windows (resampled to 0.72 s), from the **teacher** encoder.

Split — 70:30, subject-level. Subjects (not scans) are split 70/30: a subject’s scans are near-duplicates, so a scan-level split would leak identity. **TRAIN = 70 % of subjects; TEST = the held-out 30 %** (these were excluded from pretraining).

Choosing the classifier — cross-validation on TRAIN, no separate val set. The probe is an L2-regularised logistic regression; its regularisation C is selected by **subject-aware k-fold cross-validation on the TRAIN portion only** (folds are grouped by subject). The 30 % test set is **never** touched during selection. This is more robust than a fixed small val set and mirrors the LCM cross-validation protocol.

Metrics — matched to each SOTA. Most SOTA papers report **Accuracy and F1, not AUC**. We therefore report, **per task, the metric the competitor reports**: AUROC for ABIDE-Autism (vs BNT), F1 for the LCM comparisons, Accuracy for ABIDE-Age and ADNI-NC/MCI (vs SLIM-Brain / Brain-JEPA), etc. F1 is binary (positive class), as in the SOTA.

Honesty note. A near-chance task can still show an inflated F1 under class imbalance + `class_weight=balanced` (e.g. ADNI NC/MCI F1 0.66 while Acc 0.55) — for those we read the **Accuracy / AUC** as the truthful number.

4. Results per task vs SOTA

Each block lists **every SOTA that reports the same benchmark**, with **its own metric**, next to our two runs. Bold = our best run on that line.

ABIDE — Autism (psychiatric, the canonical fMRI-FM benchmark)

Model	Type	Metric	Score
BNT	supervised connectome	AUROC	0.80
BrainGFM	graph FM (25k subj)	AUC	0.71 (ABIDE II)
LCM	connectome FM	F1	0.73
Ours base	volumetric FM (~4.6k)	AUROC / F1	0.53 / 0.50
Ours fourier	volumetric FM	AUROC / F1	0.50 / 0.52

-> **Near chance, well below SOTA.** No volumetric peer reports ABIDE Autism; the models that do are larger graph/connectome models.

ABIDE — Age (demographic)

Model	Type	Metric	Score
SwiFT	volumetric FM	Acc	0.622
SLIM-Brain	volumetric FM (4.1k)	Acc	0.644
Ours base	volumetric FM (~4.6k)	Acc	0.80
Ours fourier	volumetric FM	Acc	0.71

-> **Above our direct volumetric peers.** (Caveat: our age binarisation is at the median; theirs may differ — positive but not strictly identical protocol.)

ABIDE — Sex (sanity)

Model	Metric	Score
LCM	F1	0.873
Ours base / fourier	F1	0.82 / 0.82

-> Close to LCM. (ABIDE Sex is class-imbalanced, so we read F1 as the SOTA does.)

ADNI — NC vs MCI (early Alzheimer, the main clinical duel)

Model	Type	Metric	Score
BNT	supervised	Acc	0.789
Brain-JEPA	UKB-pretrained FM	Acc / F1	0.768 / 0.863
SLIM-Brain	volumetric FM	Acc / F1	0.691 / 0.690
SwiFT	volumetric FM	Acc	0.645
Ours base	volumetric FM	Acc / F1	0.55 / 0.66
Ours fourier	volumetric FM	Acc / F1	0.54 / 0.65

-> **Near chance (Acc 0.55).** The F1 0.66 is inflated by imbalance — the Accuracy is the honest number.

ADNI — AD vs HC (full Alzheimer)

Model	Type	Metric	Score
OViTAD	supervised 2D ViT	F1	0.99 (n=31 test)
BrainGFM	graph FM	AUC / Acc	0.803 / 0.851
LCM	connectome FM	F1	0.853
Ours base	volumetric FM	AUC / Acc / F1	0.64 / 0.54 / 0.50
Ours fourier	volumetric FM	AUC / Acc / F1	0.64 / 0.61 / 0.54

-> **Modest (AUC 0.64), below SOTA.**

HCP — Sex (demographic)

Model	Type	Metric	Score
SLIM-Brain	volumetric FM	Acc / F1	0.911 / 0.911
LCM (HCP-YA)	connectome FM	F1	0.722
Ours base	volumetric FM	Acc / F1	0.81 / 0.79
Ours fourier	volumetric FM	Acc / F1	0.79 / 0.77

-> **Above LCM (F1 0.79 vs 0.72), below SLIM-Brain.**

5. Reading

The foundation model learns **global / demographic** brain structure well — it **beats the volumetric SOTA on ABIDE Age and beats LCM on HCP Sex** — but the **fine clinical signal** (autism, early Alzheimer) stays near chance, below the graph/connectome models. This is consistent with the training behaviour we diagnosed: with a mostly-frozen backbone, masking-only augmentation and 10 epochs, the encoder captures coarse structure but not subtle diagnostic patterns.

Fourier vs learned position: Fourier **did not help** — base is $\geq$ Fourier on every axis that carries signal (notably Age 0.80 vs 0.71, HCP Sex 0.81 vs 0.79). **Why this makes sense:** Fourier features pay off when coordinates are **dense and continuous** (e.g. NeRF), so the network must resolve points only a tiny distance apart. Our token grid is the opposite — only **150 discrete patches**, spaced ~ 18 mm apart and well separated. The “distinguish closely-spaced positions” problem Fourier is designed for is **barely present here**, and a 150-entry learned table already has ample capacity to place those few positions. Fourier would become relevant at a **finer resolution** (smaller patches, far more positions, or continuous voxel coordinates) — a v2 lever. The clean ablation conclusion for v1: **keep the learned positional table**.

6. Limitations and v2

- **Single 70:30 split** — headline numbers should get **mean $\pm$ std over 3–5 subject-level seeds** for publication.
- **Clinical gap** — the autism / Alzheimer gap is the target of v2.
- **v2 direction (decided):** to be **directly comparable to Brain-JEPA / BrainLM**, pretrain on a **large disjoint corpus (UK Biobank)** and evaluate on the small datasets fully held out by construction. UK Biobank access (DUA) is being applied for (2–6 months); v1 is the interim deliverable.

Cited SOTA

Brain-JEPA (NeurIPS 2024), BrainLM (ICLR 2024), SLIM-Brain (preprint 2025), BrainGFM (preprint 2025), LCM (AAAI 2026), BNT (NeurIPS 2022), SwiFT (NeurIPS 2023), OViTAD (Brain Sciences 2023). Full details in SOTA_COMPARISON_EN.